Física II – Aula 19

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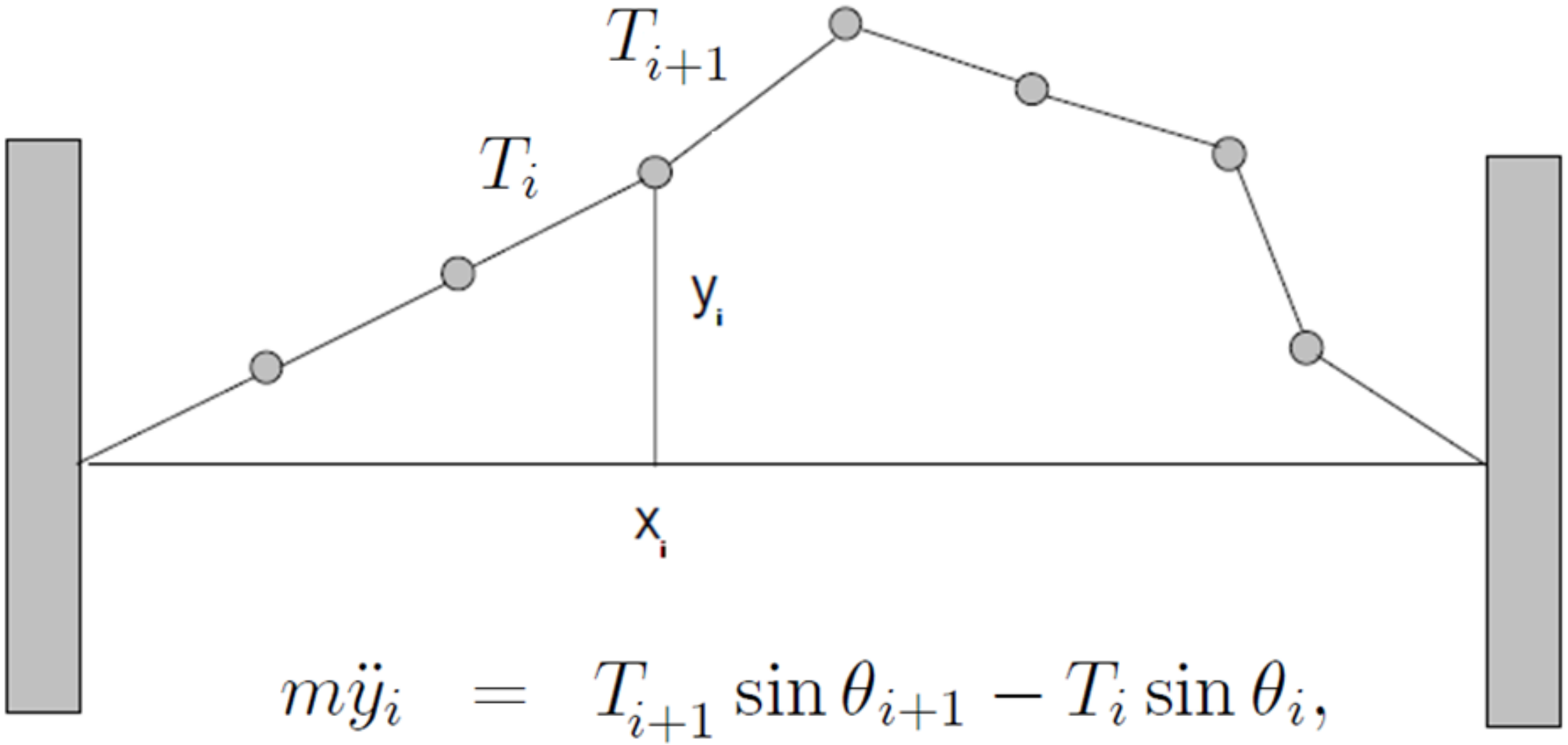
Oscilações transversais acopladas



$$
\begin{aligned}
m\ddot{y}_i &= T_{i+1}\sin\theta_{i+1} - T_i\sin\theta_i, \\
m\ddot{x}_i &= T_{i+1}\cos\theta_{i+1} - T_i\cos\theta_i
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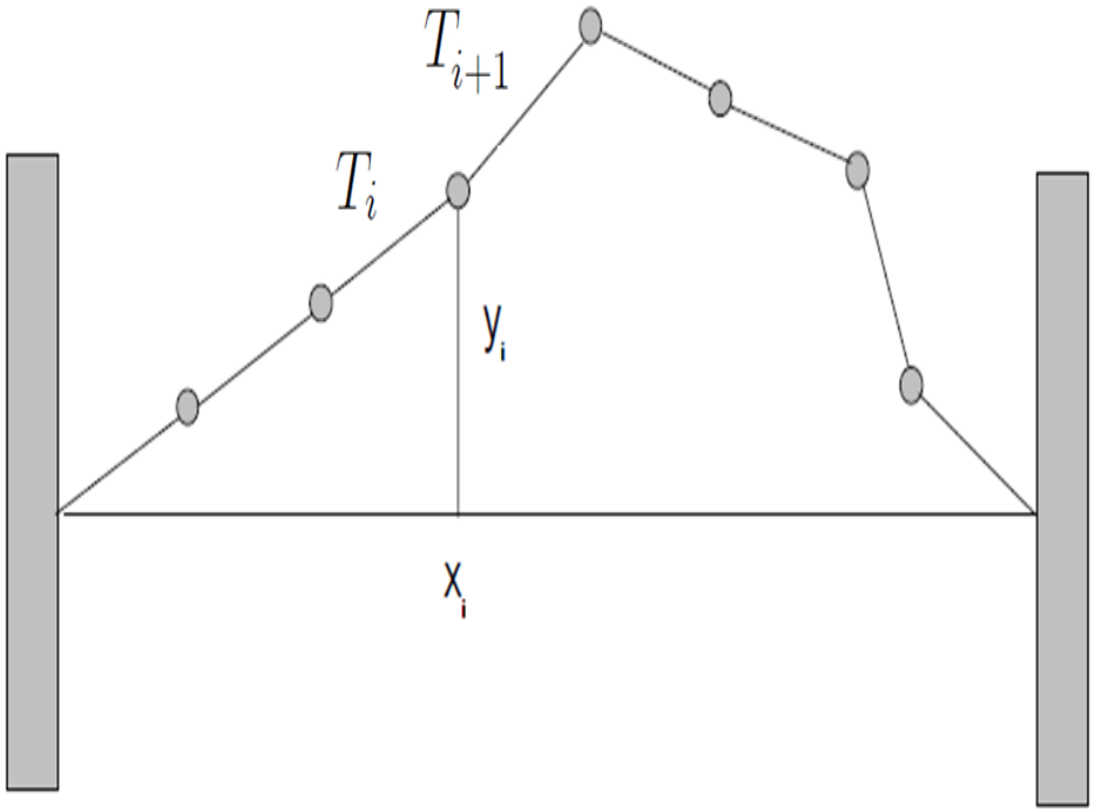
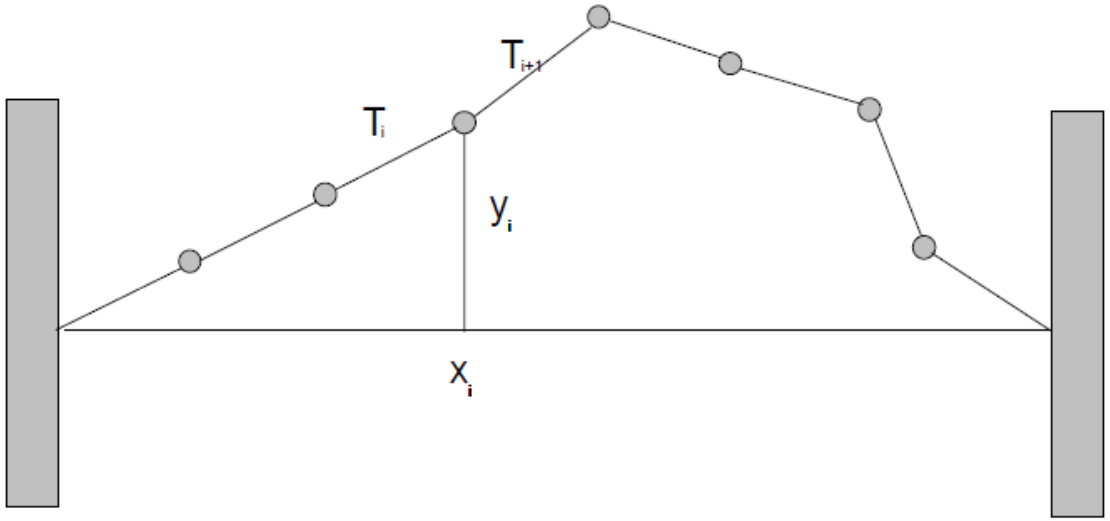


$$
\begin{aligned}
m\ddot{y}_i &= T_{i+i}\sin\theta_{i+1} - T_i\sin\theta_i, \\
0 &= T_{i+i} - T_i
\end{aligned}
$$

$$
\begin{aligned}
T_i &= \tau, \\
m\ddot{y}_i &= \tau(\sin\theta_{i+1} - \sin\theta_i) \\
&\approx \tau\left(\frac{y_{i+1} - y_i}{a} - \frac{y_i - y_{i-1}}{a}\right)
\end{aligned}
$$

$$\boxed{\ddot{y}_i = \frac{\tau}{ma}(y_{i+1} + y_{i-1} - 2y_i)}$$



$$y_0 = y_{N+1} = 0$$

$$y_J(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t)$$

$$y_J(t) = A_J \cos(\omega t)$$

$$\ddot{y}_J = -\omega^2 y_J$$

$$\boxed{\ddot{y}_i = \frac{\tau}{ma}(y_{i+1} + y_{i-1} - 2y_i)}$$

$$\boxed{\omega_0^2 = \frac{\tau}{ma}}$$

$$(-\omega^2 + 2\omega_0^2) A_1 - \omega_0^2 (A_0 + A_2) = 0$$

$$(-\omega^2 + 2\omega_0^2) A_2 - \omega_0^2 (A_1 + A_3) = 0$$

$$..$$

$$..$$

$$(-\omega^2 + 2\omega_0^2) A_J - \omega_0^2 (A_{(J-1)} + A_{(J+1)}) = 0$$

$$..$$

$$..$$

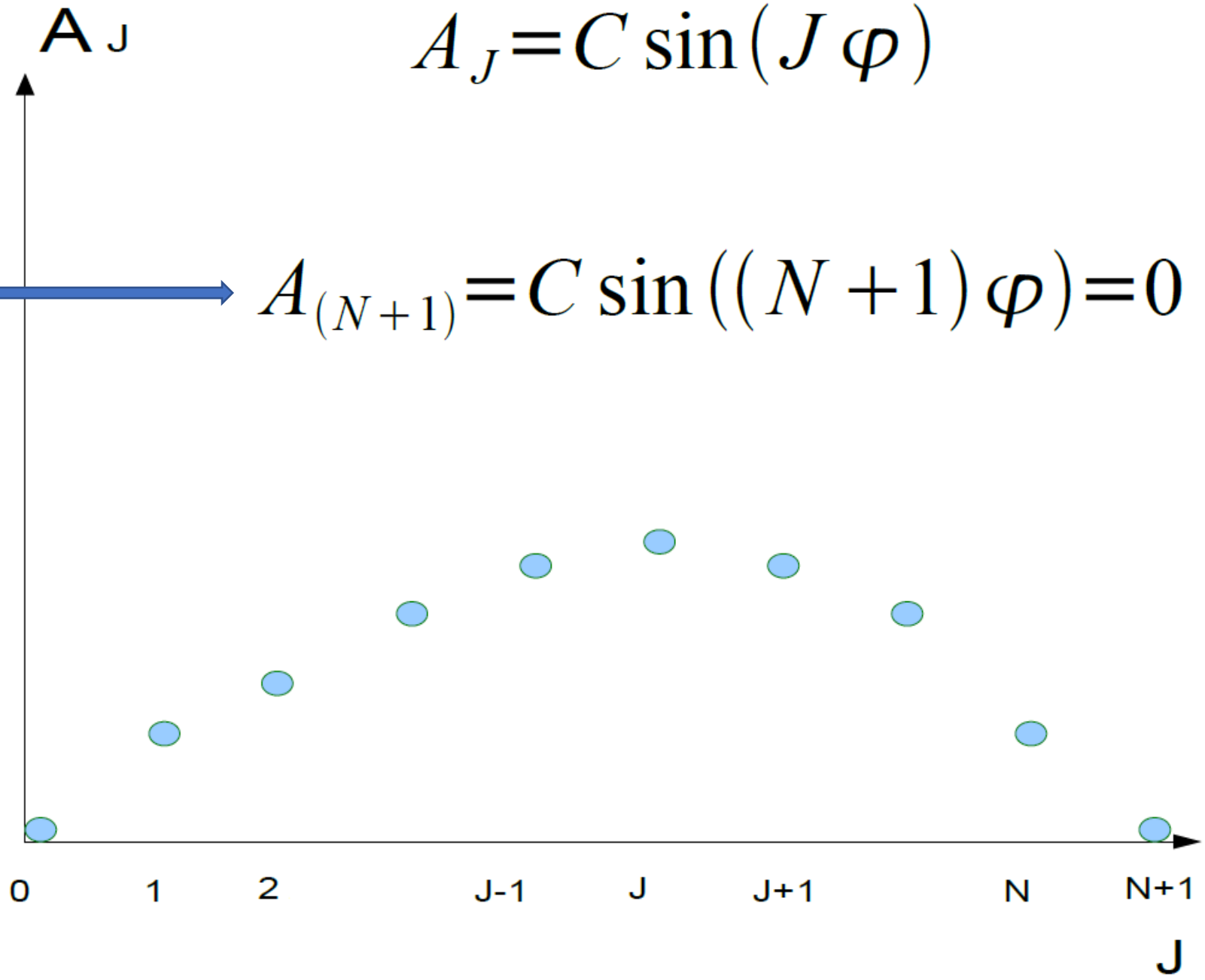
$$(-\omega^2 + 2\omega_0^2) A_N - \omega_0^2 (A_{(N-1)} + A_{(N+1)}) = 0$$

$$A_J = C\sin(J\varphi)$$

$$\underbrace{A_0 = A_{(N+1)} = 0}$$

$$A_{(N+1)} = C\sin((N+1)\varphi) = 0$$

$$A_0 = C\sin(0) = 0$$



$$A_{(N+1)} = C \sin((N+1)\varphi) = 0$$

$$(N+1)\varphi = n\pi \rightarrow n = 1, 2, \ldots$$

$$\varphi = \frac{(n\pi)}{(N+1)}$$

$$A_J = C \sin(J\varphi) \longrightarrow A_J = C \sin\left(J \frac{(n\pi)}{(N+1)}\right)$$

$$(-\omega^2+2\omega_0^2)A_J-\omega_0^2(A_{(J-1)}+A_{(J+1)})=0$$

$$\frac{(2\omega_0^2-\omega^2)}{\omega_0^2}=\frac{[A_{(J+1)}+A_{(J-1)}]}{A_J}$$

$$A_{(J-1)}=C\sin((J-1)\varphi)=c[\sin(J\varphi)\cos(\varphi)-\sin(\varphi)\cos(J\varphi)]$$
$$A_{(J+1)}=C\sin((J+1)\varphi)=c[\sin(J\varphi)\cos(\varphi)+\sin(\varphi)\cos(J\varphi)]$$

$$\frac{(2\omega_0^2-\omega^2)}{\omega_0^2}=\frac{[A_{(J+1)}+A_{(J-1)}]}{A_J}=2\cos(\varphi)=2\cos\left(\frac{(n\pi)}{(N+1)}\right)$$

$$\omega^2 = 2\omega_0^2\left[1-\cos\left(\frac{(n\pi)}{(N+1)}\right)\right]$$

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

$$2\sin^2(\theta) = 1-\cos(2\theta)$$

$$2\theta = \frac{(n\pi)}{(N+1)}$$

$$\boxed{\omega_n^2 = 4\omega_0^2\sin^2\left(\frac{(n\pi)}{(2(N+1))}\right)}$$

$$\omega_n = 2\omega_0\sin\left(\frac{(n\pi)}{(2(N+1))}\right)$$

$$A_{Jn} = C_n\sin\left(J\frac{(n\pi)}{(N+1)}\right)$$

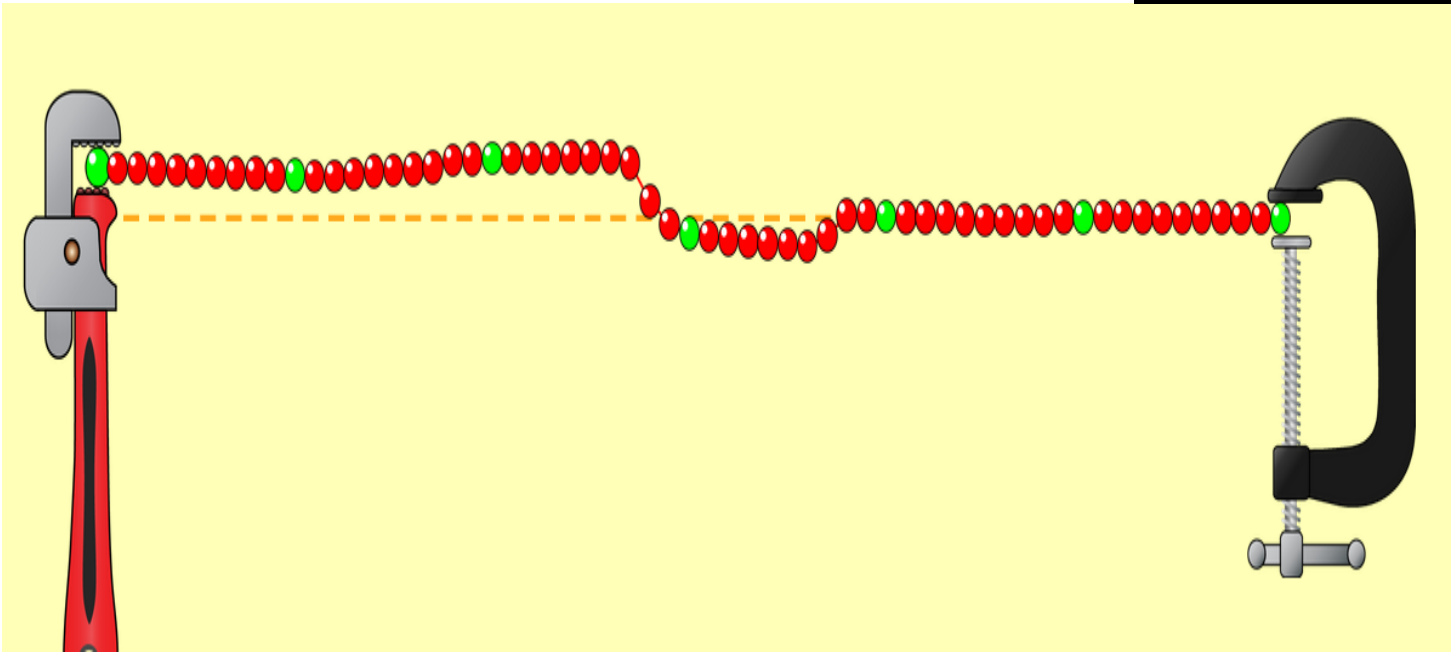
$$\left.\begin{aligned} \omega_n &= 2\omega_0 \sin\left(\frac{(n\pi)}{(2(N+1))}\right) \\ A_{Jn} &= C_n \sin\left(J\frac{(n\pi)}{(N+1)}\right) \end{aligned}\right\} \quad y_{Jn}(t) = A_{Jn}\cos(\omega_n t)$$

$$\boxed{y_{Jn}(t) = A_{Jn}\cos(\omega_n t + \delta_n)}$$

Limite para o continuum

$$N \to \infty,\ a \to 0$$

$$\frac{m}{a} \to \sigma$$



$$\ddot{y}_i = \frac{\tau}{ma}(y_{i+1} + y_{i-1} - 2y_i)$$

$$y_i \to y(x,t)$$

https://phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_pt_BR.html

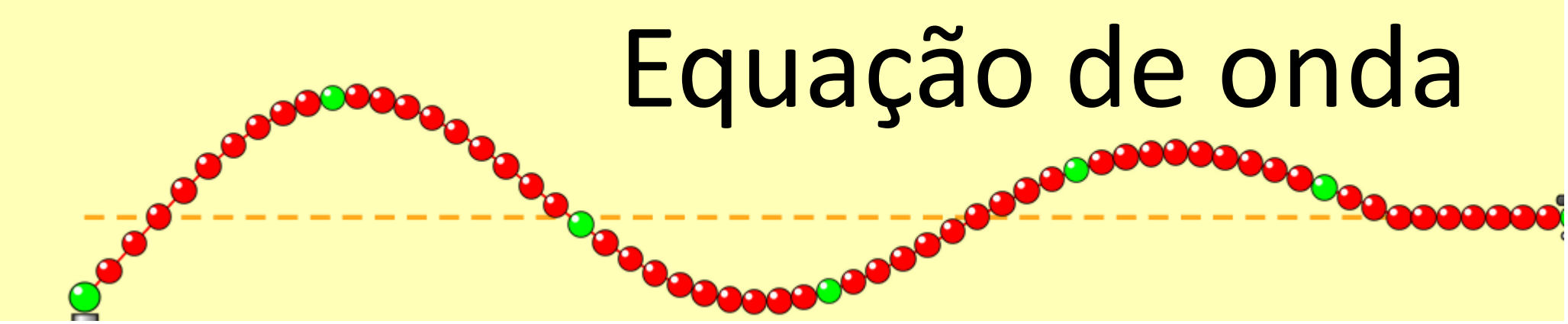
$$y_{i+1} \rightarrow y(x+a,t) = y(x,t) + \frac{\partial y(x,t)}{\partial x} a + \frac{1}{2}\frac{\partial^2 y(x,t)}{\partial x^2} a^2,$$

$$y_{i-1} \rightarrow y(x-a,t) = y(x,t) - \frac{\partial y(x,t)}{\partial x} a + \frac{1}{2}\frac{\partial^2 y(x,t)}{\partial x^2} a^2,$$

$$\ddot{y}_i = \frac{\tau}{ma}(y_{i+1} + y_{i-1} - 2y_i)$$

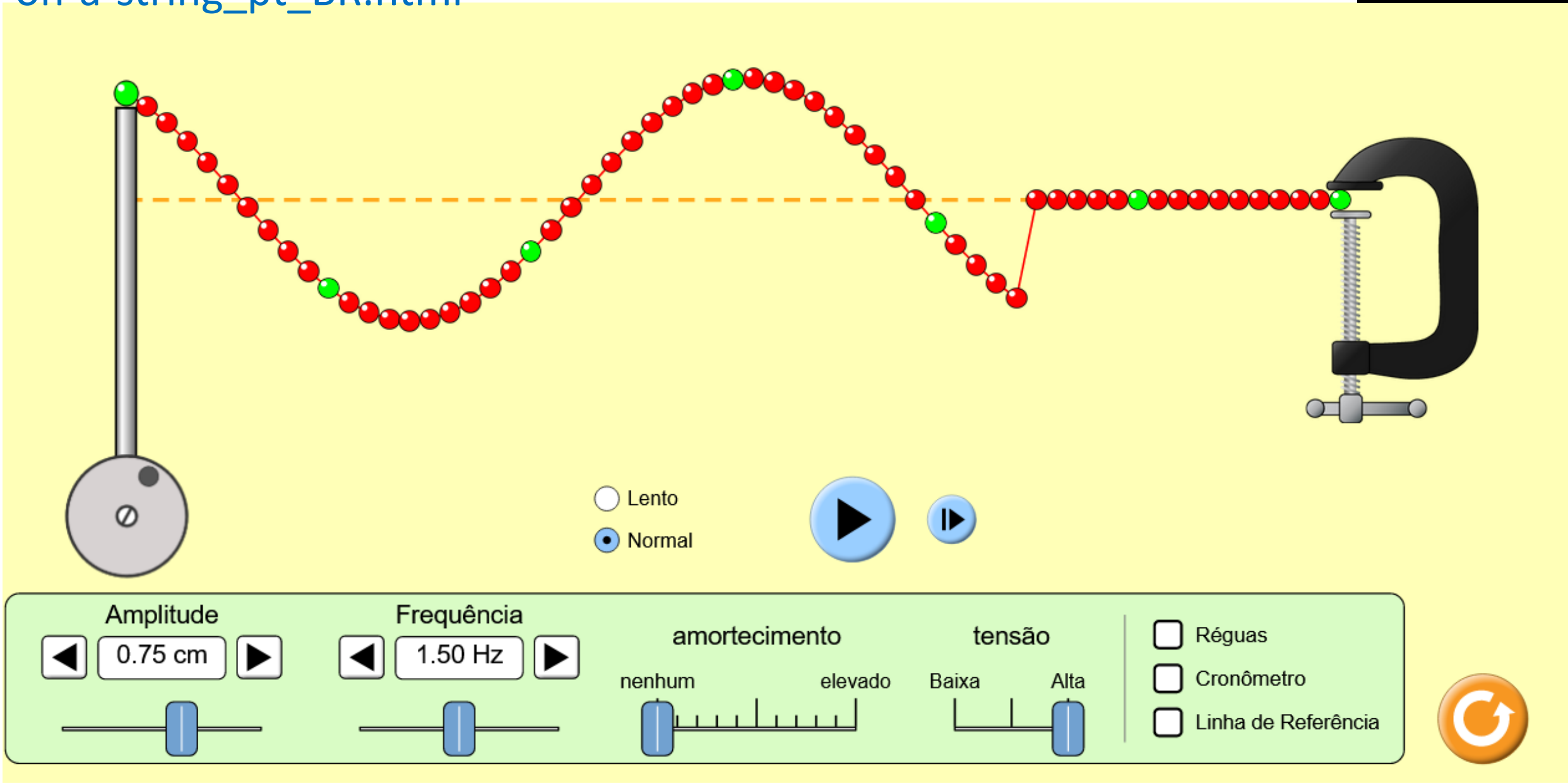
$$\begin{aligned} \frac{\partial^2 y(x,t)}{\partial t^2} &= \frac{\tau}{ma}\frac{\partial^2 y(x,t)}{\partial x^2} a^2 \\ &= \frac{a\tau}{m}\frac{\partial^2 y(x,t)}{\partial x^2}, \end{aligned}$$

Equação de onda



$$\frac{\partial^2 y(x,t)}{\partial t^2} - \frac{\tau}{\sigma}\frac{\partial^2 y(x,t)}{\partial x^2} = 0.$$

$$\frac{\partial^2 y(x,t)}{\partial t^2} - v^2\frac{\partial^2 y(x,t)}{\partial x^2} = 0.$$

$$v = \sqrt{\frac{\tau}{\sigma}},$$

https://phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_pt_BR.html